

# Analyzing Multicollinearity in Octo-Bit Patterns.

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## 1 Appendix C. Heatmap.

This analysis highlights a critical issue of multicollinearity among the predictor variables, which occurs when independent variables are highly correlated. Multicollinearity complicates the ability of traditional regression models to estimate the individual effects of each predictor, resulting in unstable and unreliable coefficient estimates. To illustrate this, Figure 1 shows this problem.

The heatmap in Figure 1 visualizes the correlation matrix of the octo-bit patterns, where each cell represents the correlation coefficient between two patterns. The color scale ranges from blue (negative correlation) to red (positive correlation), with white representing near-zero correlation.

## 2 Key Observations

- The presence of strong red blocks along the diagonal suggests high multicollinearity among subsets of patterns.
- Alternating blue and red patterns indicate the existence of complex dependencies rather than simple pairwise correlations.
- The structured nature of the heatmap suggests systematic dependencies between certain groups of octo-bit patterns.

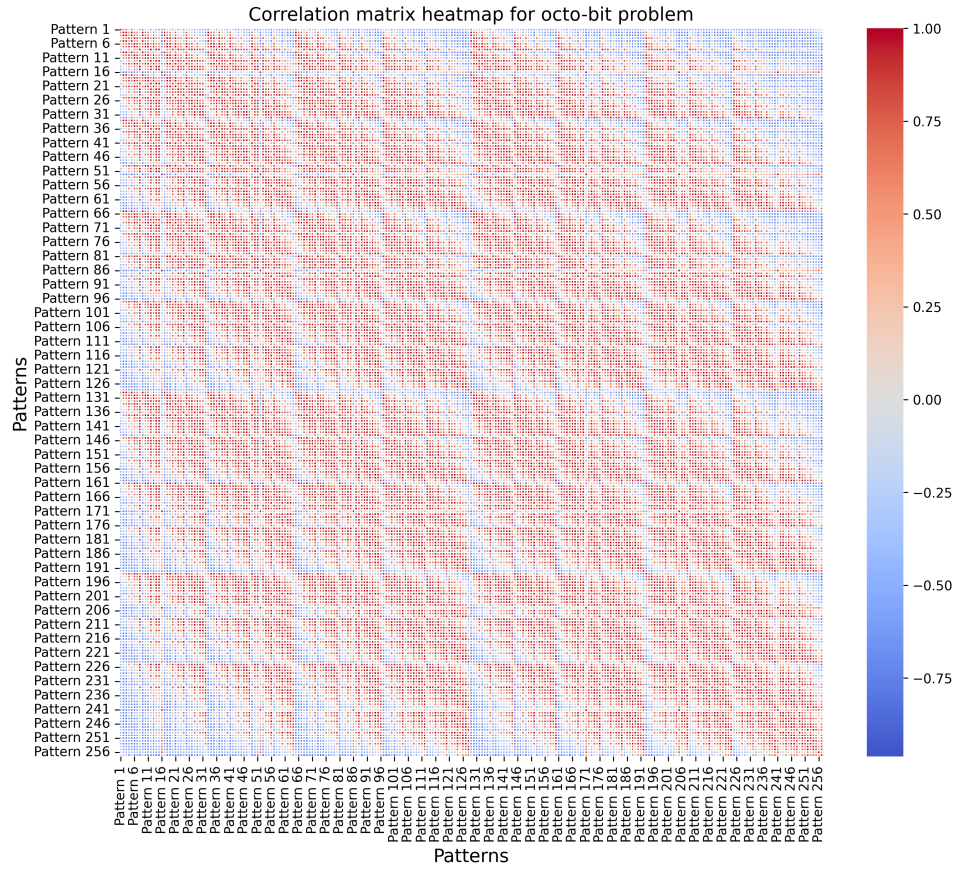


Figure 1: Heatmap of the octo-bits multicollinearity problem